[Assignment-4]

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Section : 28

Course code : CSE423

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Ans to the Question no 1

a) The HSL and HSV color models are both cylindrical representations of color but differ fundamentally in how they perceive brightness. In HSL, lightness represents the perceived brightness where L = 0 is black, L = 1 is white, and pure hues are at L = 0.5. This model is intuitive for human perception, making it suitable for color adjustment in digital imaging tools. In contrast, HSV uses value to represent brightness where V = 0 is black and V = 1 is the brightest color, with saturation describing how much white is added. HSV alligns better with low light works physically, so it is commonly used is computer graphics for shading and lighting, as well as in user interface design for color pickers where brightness adjustment is key.

b) HSV $(45°, 0.8, 0.9)$

Here, $C_{max} = R = V = 0.9$

$$S = 1 - \frac{C_{min}}{C_{max}}$$

$$\Rightarrow 0.8 = 1 - \frac{C_{min}}{0.9}$$

$$\therefore C_{min} = 0.18$$

$C_{max} = R$,

$$H = \frac{g - b}{C_{max} - C_{min}} \times 60$$

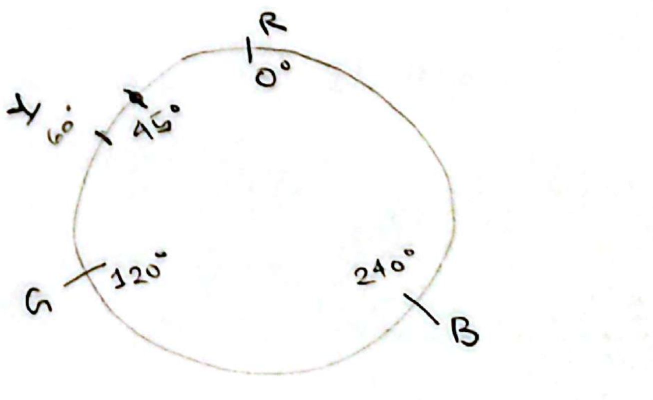
$$\Rightarrow 45 = \frac{g - 0.18}{0.9 - 0.18} \times 60$$

$$\therefore g = 0.72$$

$\therefore$ RGB $(0.9, 0.72, 0.18)$

$\therefore$ cmy $(1-0.9, 1-0.72, 1-0.18)$

$= (0.1, 0.28, 0.82)$



Ans to the question no 2

a)

Light 1 :

Position $L1 = (5, 10, 7)$

Intensity $I1 = 20$

Radius $R1 = 25$

Light 2 :

Position $L2 = (15, 30, 45)$

Intensity $I2 = 10$

Radius $R2 = 60$

L1:

Attenuation factors:

$$f_{att_1} = \max\left(1 - \left(\frac{d_1}{r_1}\right)^2, 0\right)$$

$$\cancel{f_{att_1}} = \max\left(1 - \left(\frac{d_1}{r_1}\right)^2, 0\right)$$

$$= \max\left(1 - \left(\frac{7\sqrt{2}}{25}\right)^2, 0\right)$$

$$= 0.8432$$

$$f_{att_2} = \max\left(1 - \left(\frac{d_2}{r_2}\right)^2, 0\right)$$

$$= \max\left(1 - \left(\frac{51.87}{60}\right)^2, 0\right)$$

$$= 0.2526$$

Given point, $P = (0, 2, 4)$

xz plane,

So normal vector $N = (0, 1, 0)$

Camera position, V $(3, 2, 6)$

Specular coefficients : $K_a = 0.2$

$K_d = 0.6$

$K_s = 0.4$

Shininess factor, $n = 10$

Ambient light intensity:

I ambient $= 8$

$$d_1 = \sqrt{(5-0)^2 + (10-2)^2 + (7-4)^2}$$

$$= \sqrt{5^2 + 8^2 + 3^2}$$

$$= 7\sqrt{2}$$

$$d_2 = \sqrt{(15-0)^2 + (30-2)^2 + (45-4)^2}$$

$$= \sqrt{15^2 + 28^2 + 41^2}$$

$$\approx 51.87$$

$$\vec{L}_1 = L_1 - P$$
$$= (5,10,7) - (0,2,4)$$
$$= (5,8,3)$$

$$\hat{n}_1 = \hat{j}$$

$$\hat{L}_1 = \frac{5\hat{i} + 8\hat{j} + 3\hat{k}}{\sqrt{5^2 + 8^2 + 3^2}}$$
$$= \left(\frac{5}{7\sqrt{2}}\right)\hat{i} + \left(\frac{8}{7\sqrt{2}}\right)\hat{j} + \left(\frac{3}{7\sqrt{2}}\right)\hat{k}$$

$$\hat{R}_1 = 2(\hat{L}_1 \cdot \hat{n}_1)\hat{n}_1 - \hat{L}_1$$
$$= 2\left(\frac{8}{7\sqrt{2}}\right)\hat{j} - \hat{L}_1$$
$$= \frac{16}{7\sqrt{2}}\hat{j} - \frac{5}{7\sqrt{2}}\hat{i} - \frac{8}{7\sqrt{2}}\hat{j} - \frac{3}{7\sqrt{2}}\hat{k}$$
$$= -\frac{5}{7\sqrt{2}}\hat{i} + \frac{8}{7\sqrt{2}}\hat{j} - \frac{3}{7\sqrt{2}}\hat{k}$$

$$\vec{V} = V - P$$
$$= (3,2,6) - (0,2,4)$$
$$= (3,0,2)$$
$$\hat{V} = \frac{3\hat{i} + 2\hat{k}}{\sqrt{3^2 + 2^2}} = \frac{3}{\sqrt{13}}\hat{i} + \frac{2}{\sqrt{13}}\hat{k}$$

$$\hat{R}_1 \cdot \hat{V} = \left(-\frac{5}{7\sqrt{2}}\hat{i} + \frac{8}{7\sqrt{2}}\hat{j} - \frac{3}{7\sqrt{2}}\hat{k}\right) \cdot \left(\frac{3}{\sqrt{13}}\hat{i} + \frac{2}{\sqrt{13}}\hat{k}\right)$$
$$= -0.42 - 0.17$$
$$= -0.59$$

$$\vec{L}_2 = L_2 - P$$
$$= (15,30,45) - (0,2,4)$$
$$= (15,28,41)$$

$$\hat{n} = \hat{j}$$

$$\hat{L}_2 = \frac{15\hat{i} + 28\hat{j} + 41\hat{k}}{\sqrt{15^2 + 28^2 + 41^2}}$$
$$= \left(\frac{15}{51.87}\right)\hat{i} + \left(\frac{28}{51.87}\right)\hat{j} + \left(\frac{41}{51.87}\right)\hat{k}$$

$$\hat{R}_2 = 2(\hat{L}_2 \cdot \hat{n}_2)\hat{n}_2 - \hat{L}_2$$
$$= 2 \times \left(\frac{28}{51.87}\right)\hat{j} - \hat{L}_2$$
$$= \frac{56}{51.87}\hat{j} - \frac{15}{51.87}\hat{i} - \frac{28}{51.87}\hat{j} - \frac{41}{51.87}\hat{k}$$
$$= -\frac{15}{51.87}\hat{i} + \frac{28}{51.87}\hat{j} - \frac{41}{51.87}\hat{k}$$

~~$\vec{V} = V - P$~~

~~$= ($~~

$$\hat{R}_2 \cdot \hat{V} = \left(-\frac{15}{51.87}\hat{i} + \frac{28}{51.87}\hat{j} - \frac{41}{51.87}\hat{k}\right) \cdot \left(\frac{3}{\sqrt{13}}\hat{i} + \frac{2}{\sqrt{13}}\hat{k}\right)$$
$$= -0.24 - 0.44$$
$$= -0.68$$

$$(\hat{R}_1 \hat{V})^{10} = (-0.59)^{10}$$

$$= \cancel{0.3481}\ 5.11 \times 10^{-3}$$

$$(\hat{R}_2 \hat{V})^{10} = (-0.68)^{10}$$

$$= 0.02$$

$$\therefore I_1 = 20 \times \cancel{0.2}\ 0.8432 \left[0.6 \times \frac{8}{7\sqrt{2}} + 0.4 \times \overset{5.11\times10^{-3}}{\cancel{0.3481}} \right]$$

$$= \cancel{10.53}\ \cancel{0.4869}\ 8.21$$

$$I_2 = 10 \times 0.2526 \left[0.6 \times \frac{28}{51.87} + 0.4 \times 0.02 \right]$$

$$= 0.84$$

so,

$$I = I_a k_a + I_1 + I_2$$

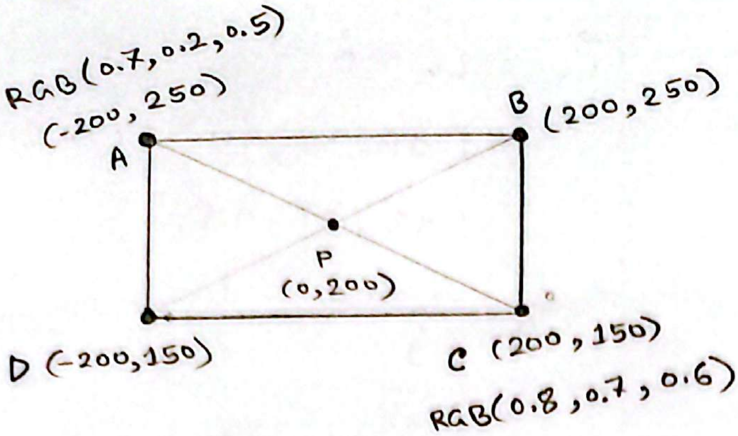
$$= 8 \times 0.2 + \overset{8.21}{\cancel{10.53}} + 0.84$$

$$\cancel{0.4869}$$

$$= \cancel{12.97}\ \cancel{2.9269}\ 10.65$$

(Ans)

b)



$$P = \frac{\cancel{-200}+\cancel{200}+\cancel{200}-\cancel{200}}{4}, \frac{250+250+150+150}{4}$$

$$= \left(\frac{0}{4}, 200\right)$$

$$= (0, 200)$$

color at A : RGB (0.7, 0.2, 0.5)

color at C : CMY (0.2, 0.3, 0.4)

RGB (0.8, 0.7, 0.6)

For Gouraud scaling,

Now,

$$color_p = color_A + \frac{distance_{AP}}{distance_{AC}} \times (color_C - color_A)$$

$$= (0.7, 0.2, 0.5) + \frac{200}{400} \times (0.1, 0.5, 0.1)$$

$$= (0.7, 0.2, 0.5 + \frac{\cancel{200}}{\cancel{400}} \times (0.05, 0.25, 0.05)$$

$$= (0.75, 0.45, 0.55)$$

$\therefore$ RGB(0.75, 0.45, 0.55)